Programming Assignment Report

Design and Analysis of Algorithms:

Name: ASHISH BEHERA

Registration Number: 2411021060108 **Problem**

Name: COMPARE THE TRIPLETS

Alice and Bob each created one problem for HackerRank. A reviewer rates the two challenges, awarding points on a scale from 1 to 100 for three categories: problem clarity, originality, and difficulty. The rating for Alice's challenge is the triplet a = (a[0], a[1], a[2]), and the rating for Bob's challenge is the triplet b = (b[0], b[1], b[2]). The task is to calculate their comparison points by comparing each category: If a[i] > b[i], then Alice is awarded 1 point. If a[i] < b[i], then Bob is awarded 1 point. If a[i] = b[i], then neither person receives a point. Example a = [1, 2, 3] b = [3, 2, 1] For elements *0*, Bob is awarded a point because a[0] < b[0]. For the equal elements a[1] and b[1], no points are earned. Finally, for elements 2, a[2] > b[2] so Alice receives a point.

Problem Hacker Rank Link: https://www.hackerrank.com/challenges/compare-the-triplets/problem

Solution Steps

1. Read Input

- Read three integers **i, j, and k**.
- These represent: ○ **i** → starting day ○ **j** → ending day
    - ○ **k** → divisor

2. Initialize Counter

> Create a variable **count**.
> This will store the number of beautiful days.

3. Iterate Through Days

> Loop through each day from **i to j**.

for (day = i → j)

4. Reverse the Number

For each day:

>> Extract digits using modulus %

>>Build the reverse number.

Example: num = day

reverse = reverse * 10 + num % 10

num = num / 10

5. Calculate Absolute Difference

Compute: diff = |day - reverse|

6. Check Divisibility

Check if:

diff % k == 0

If true → the day is **beautiful**.

7. Increment Counter

> If the condition is satisfied:

count++

8. Return Result

> After checking all days:

> Return the total number of beautiful days**.**

Complete JAVA Code:

```
#include <bits/stdc++.h>

using namespace std;

string ltrim(const string &);
string rtrim(const string &);
vector<string> split(const string &);

/*
 * Complete the 'compareTriplets' function below.
 *
 * The function is expected to return an INTEGER_ARRAY.
 * The function accepts following parameters:
 *  1. INTEGER_ARRAY a
 *  2. INTEGER_ARRAY b
 */

vector<int> compareTriplets(vector<int> a, vector<int> b) {

}

int main()
```

```
{
    ofstream fout(getenv("OUTPUT_PATH"));

    string a_temp_temp;
    getline(cin, a_temp_temp);

    vector<string> a_temp = split(rtrim(a_temp_temp));

    vector<int> a(3);

    for (int i = 0; i < 3; i++) {
        int a_item = stoi(a_temp[i]);

        a[i] = a_item;
    }

    string b_temp_temp;
    getline(cin, b_temp_temp);

    vector<string> b_temp = split(rtrim(b_temp_temp));

    vector<int> b(3);
```

```
for (int i = 0; i < 3; i++) {
    int b_item = stoi(b_temp[i]);

    b[i] = b_item;
}

vector<int> result = compareTriplets(a, b);

for (size_t i = 0; i < result.size(); i++) {
    fout << result[i];

    if (i != result.size() - 1) {
        fout << " ";
    }
}

fout << "\n";

fout.close();

return 0;
```

```
}

string ltrim(const string &str) {
    string s(str);

    s.erase(
        s.begin(),
        find_if(s.begin(), s.end(), not1(ptr_fun<int, int>(isspace)))
    );

    return s;
}

string rtrim(const string &str) {
    string s(str);

    s.erase(
        find_if(s.rbegin(), s.rend(), not1(ptr_fun<int, int>(isspace))).base(),
        s.end()
    );

    return s;
```

```
}

vector<string> split(const string &str) {
    vector<string> tokens;

    string::size_type start = 0;
    string::size_type end = 0;

    while ((end = str.find(" ", start)) != string::npos) {
        tokens.push_back(str.substr(start, end - start));

        start = end + 1;
    }

    tokens.push_back(str.substr(start));

    return tokens;
}
```

Sample Input:

20 23 6

Sample Output:

2

Observation / Solution :

Key Idea

The main concept is to compare a number with its reverse and determine whether their difference is divisible by k.

Approach

1. Iterate through all numbers in the range i to j.
2. Reverse each number.
3. Calculate the absolute difference.
4. Check if the difference is divisible by k.
5. Count such numbers**.**

Complexity:

Time Complexity:

$$O(n \times d)$$

Where:

- **n** = number of days
- **d** = number of digits

Since digits are small, this is effectively:

$$O(n)$$

Space Complexity:

$$O(1)$$

Only a few variables are used.